

# CUSTOMER CHURN ANALYTICS

## Dashboard · Task 2 Insight Report

Digital Kuppam | Data Science Internship

|           |                                                    |
|-----------|----------------------------------------------------|
| Project   | Customer Churn Analytics Dashboard                 |
| Dataset   | Customer Churn Dataset                             |
| Records   | 7,043 customers                                    |
| Target    | Churn (Yes / No)                                   |
| Dashboard | Python + Streamlit                                 |
| Focus     | Churn patterns, segmentation and retention actions |

## 1. Introduction

Customer churn is an important issue for a telecommunications business because every customer who leaves can reduce recurring revenue and increase the effort required to acquire a replacement. The purpose of this project is to study customer-level churn data, identify the groups that are more likely to leave, and convert those patterns into practical retention actions.

The analysis was completed as an exploratory business analytics exercise. The dashboard brings together key numbers, comparisons and filters so that churn can be viewed from different customer and service dimensions instead of relying on a single overall percentage.

**Main objective:** understand where churn is concentrated and provide recommendations that a business team could realistically test.

## 2. Dataset Description

The dataset contains 7,043 customer records and 21 original columns. The target variable is **Churn**, which records whether a customer left the company.

| Area             | Examples / Description                                                      |
|------------------|-----------------------------------------------------------------------------|
| Customer details | customerID, gender, SeniorCitizen, Partner, Dependents                      |
| Tenure           | tenure – customer relationship length in months                             |
| Services         | Phone, internet, security, backup, device protection, support and streaming |
| Billing          | Contract, PaperlessBilling, PaymentMethod                                   |
| Charges          | MonthlyCharges, TotalCharges                                                |
| Target           | Churn – Yes / No                                                            |

The dataset gives enough information to compare churn across customer characteristics, contract commitment, service usage and payment behaviour.

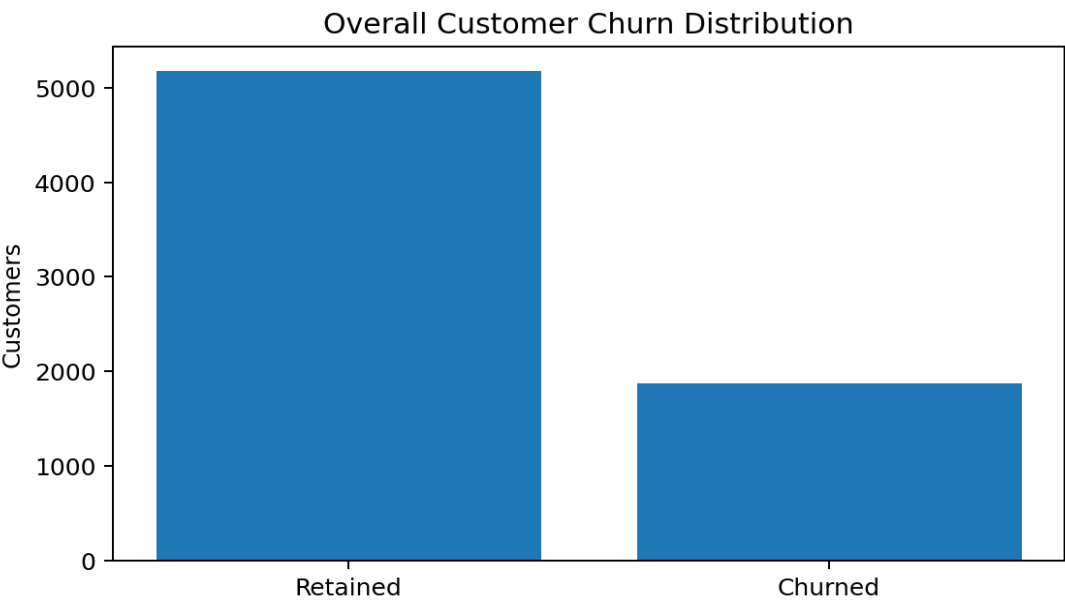
### 3. Data Preparation

| Step | Action Taken                                                      | Reason                                   |
|------|-------------------------------------------------------------------|------------------------------------------|
| 1    | Converted TotalCharges from text to numeric                       | Allows charge-based calculations         |
| 2    | Handled 11 blank/non-numeric TotalCharges values                  | Avoids errors in numerical analysis      |
| 3    | Removed customerID from analytical calculations                   | Identifier has no useful business signal |
| 4    | Standardized Churn values for analysis                            | Makes churn-rate calculations consistent |
| 5    | Separated categorical and numerical variables                     | Makes analysis and visualization easier  |
| 6    | Created derived variables such as TenureYears and AvgMonthlySpend | Adds useful business context             |

After preparation, the analysis focused on reliable comparisons rather than simply displaying raw fields. Charge-dependent calculations were based on usable numeric values, while the original customer dataset remained unchanged.

### 4. Key Metrics

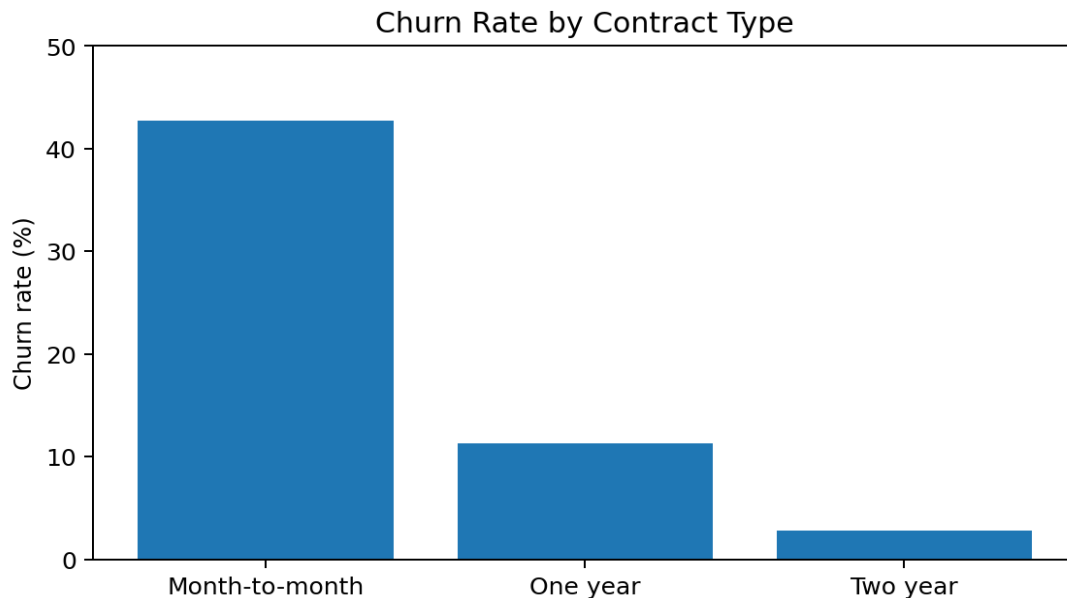
| Metric                  | Result       | Interpretation                          |
|-------------------------|--------------|-----------------------------------------|
| Total customers         | 7,043        | Customer records in the dataset         |
| Churned customers       | 1,869        | Customers who left                      |
| Retained customers      | 5,174        | Customers who stayed                    |
| Overall churn rate      | 26.54%       | More than one in four customers churned |
| Average tenure          | 32.37 months | Average customer relationship length    |
| Average monthly charges | \$64.76      | Average monthly customer charge         |



## 5. Visualizations

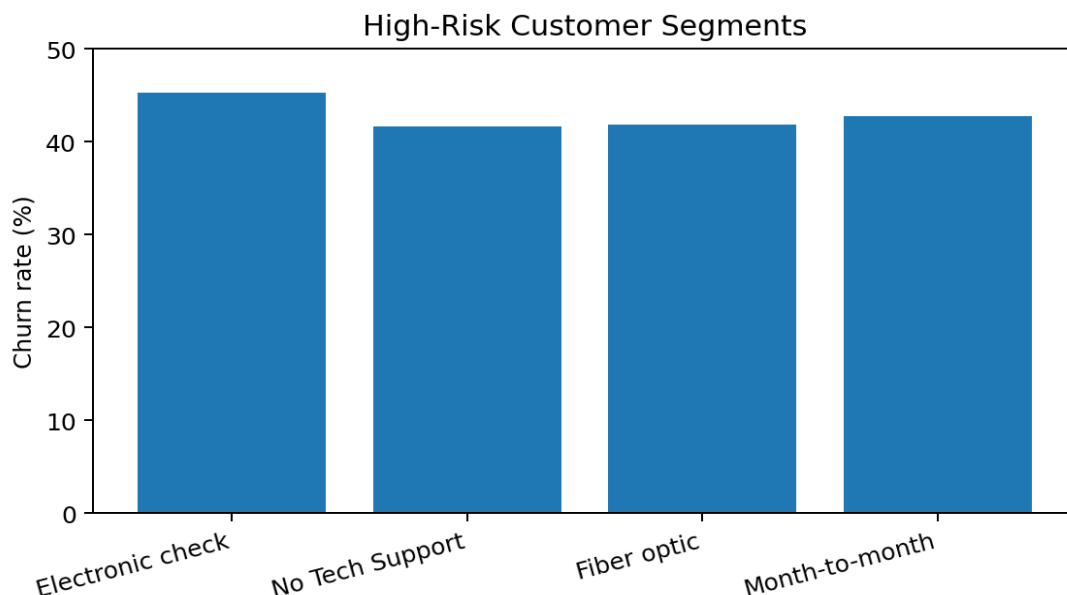
The dashboard uses visual comparisons to make differences between customer groups easier to identify. The following views were especially useful for the analysis.

### 5.1 Churn by Contract Type



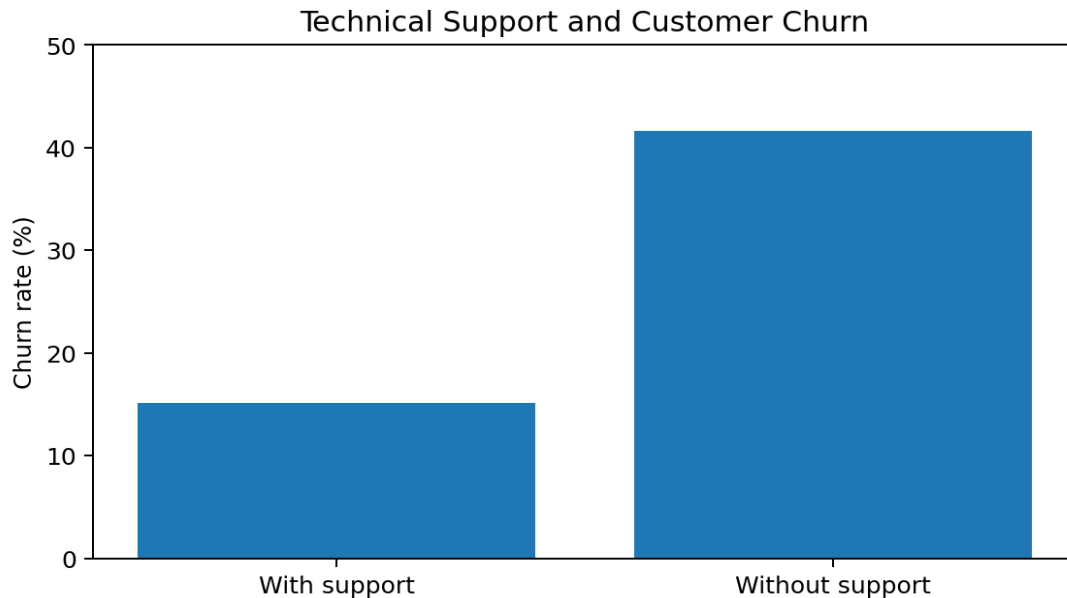
Month-to-month customers have a 42.71% churn rate, compared with 11.27% for one-year contracts and 2.83% for two-year contracts. The gap is large enough to make contract commitment one of the clearest segments to focus on.

### 5.2 Other High-Risk Segments



Electronic-check users show a 45.29% churn rate. Fiber-optic customers show 41.89%, customers without technical support show 41.64%, and month-to-month customers show 42.71%. These figures point to several overlapping areas where retention efforts can be concentrated.

## 5.3 Technical Support Comparison



Customers without technical support have a 41.64% churn rate, while customers with technical support have a 15.17% churn rate. This does not by itself prove that support causes lower churn, but it is a strong association worth testing through targeted service and retention programs.

## 5.4 Customer Distribution by Gender

The customer base is broadly balanced, with 3,488 female customers and 3,555 male customers. This suggests that gender is not the main story in the overall customer mix; the stronger business differences appear in contract, tenure, charges, payment and service-related segments.

## 6. Key Findings

1. **Overall churn:** 1,869 of 7,043 customers churned, giving an overall churn rate of 26.54%.
2. **Contract:** Month-to-month customers are substantially more likely to churn than customers on longer contracts.
3. **Tenure:** Churned customers average about 17.98 months of tenure, compared with 37.57 months for retained customers.
4. **Charges:** Churned customers average \$74.44 per month, compared with \$61.27 for retained customers.
5. **Internet service:** Fiber-optic customers have a 41.89% churn rate, higher than DSL and no-internet groups.
6. **Payment method:** Electronic-check customers have the highest payment-group churn rate at 45.29%.
7. **Support:** Customers without technical support have much higher churn than customers with support.

## 7. Five Business Insights

### 1. Month-to-month customers are the highest-risk segment

A 42.71% churn rate shows that customers with low contract commitment need more attention. Moving suitable customers toward longer commitments could improve retention.

### 2. The early customer period is important

Churned customers have a much shorter average tenure than retained customers. The first few months are therefore a practical point for stronger onboarding and proactive engagement.

### 3. Higher monthly charges may indicate value concerns

The average monthly charge is higher for churned customers (\$74.44) than for retained customers (\$61.27). This may reflect price sensitivity, service expectations or a mismatch between cost and perceived value.

### 4. Electronic-check users form a high-risk payment segment

The 45.29% churn rate for electronic-check users suggests that the payment journey deserves attention. Billing clarity, convenience and payment migration can be tested.

### 5. Technical support is associated with lower churn

The difference between 15.17% churn with support and 41.64% without support is substantial. Support and service assistance may be useful components of a retention strategy.

## 8. Recommendations

- Contract conversion:** Offer suitable loyalty benefits or incentives to high-risk month-to-month customers to encourage longer contracts.
- Early-life retention:** Strengthen onboarding, customer education and proactive support during the first 3–6 months.
- Support and security:** Proactively offer technical support and online-security services to customers who currently lack them.
- Pricing and value review:** Investigate customers with high monthly charges and test personalized plans or retention offers.
- Payment improvement:** Encourage convenient automatic payment options and investigate friction around electronic checks.
- Targeted campaigns:** Use dashboard filters and, where available, churn probability to prioritize customers instead of contacting everyone equally.

## 9. Conclusion

The Customer Churn Analytics Dashboard provides a clear view of where customer churn is concentrated across the 7,043-customer dataset. The overall churn rate of 26.54% shows that retention is an important business area, but the analysis also shows that churn is not evenly distributed across all customers.

The strongest differences are seen across contract type, tenure, monthly charges, internet service, payment method and technical support. Month-to-month and shorter-tenure customers represent important groups for early retention efforts, while higher-charge, electronic-check and customers without technical support also show comparatively higher churn.

These findings can help the business move from a general retention approach toward more focused actions such as stronger onboarding, suitable contract offers, better support coverage, payment improvements and targeted retention campaigns. The results should be treated as useful business signals rather than proof of cause and effect. The next step is to test these actions and measure churn reduction, retention performance and return on investment over time.